AUTOMATING THE CLUSTERING OF ENVIRONMENTAL GENOMES

Exploring approaches based on Answer Set Programming

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INTRODUCTION

CONTEXT AND CHALLENGES

This project is part of the work of the **LAGE** (Genoscope), which explores the diversity of marine **eukaryotic** communities using omics data.

As part of the **Tara Oceans** project, plankton samples were collected worldwide, sequenced, and then assembled into **contigs**. The goal: **reconstruct MAGs** (*Metagenome-Assembled Genomes*) by grouping contigs that come from the same organism (binning).

Until now, MAGs were reconstructed manually using the **Anvi'o software**, which displays how the contigs are distributed. The objective is **to automate this task**.

OUR APPROACH

INTRODUCTION TO ANSWER SET PROGRAMMING

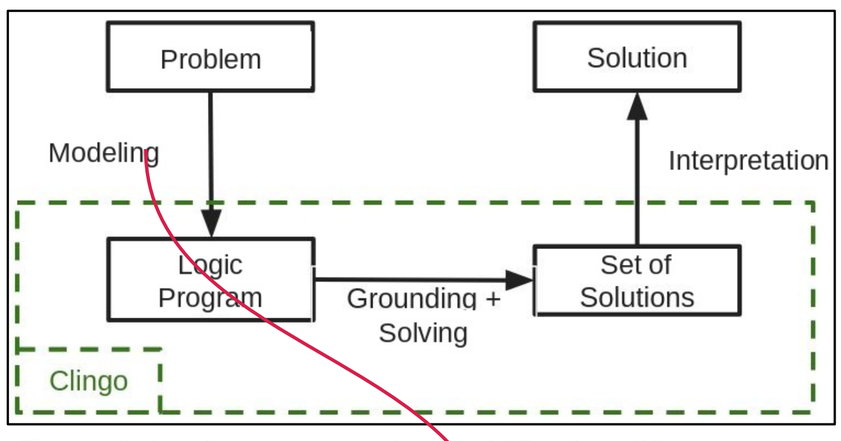


Figure 3: Logic programming in ASP using Clingo

OUR WORKFLOW

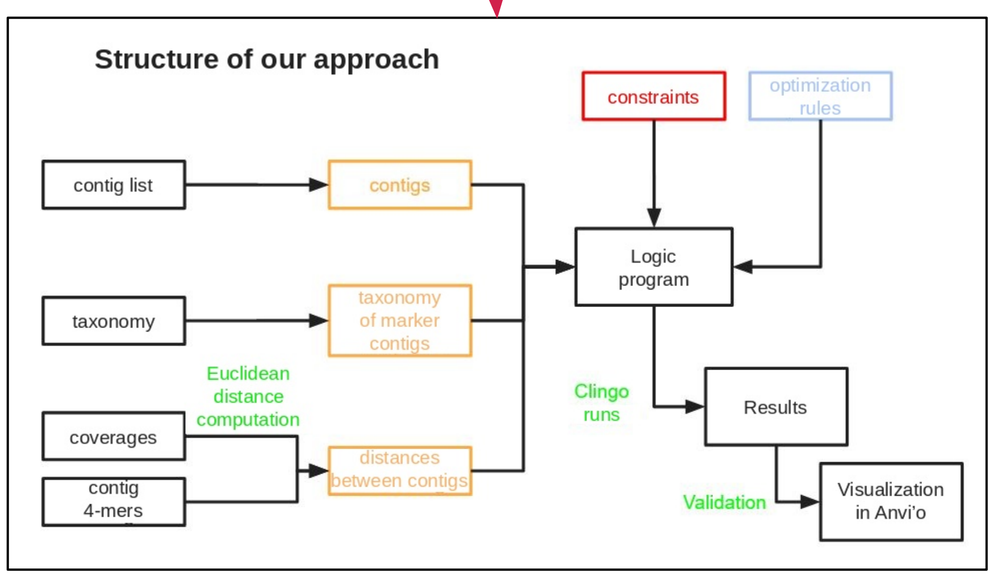


Figure 4: Structure of our automated approach

CONSTRAINTS AND OPTIMIZATION RULES

Constraints: each contig is assigned to exactly one cluster, and clusters containing contigs that differ up to taxonomic rank 3 are forbidden.

Optimization rules:

- Maximize taxonomic consistency
- Maximize inter-cluster distance
- Maximize completeness (number of distinct genes per cluster)
- Minimize intra-cluster distances
- Minimize gene redundancy (multiple occurrences of the same gene within a cluster)

MAG reconstruction process

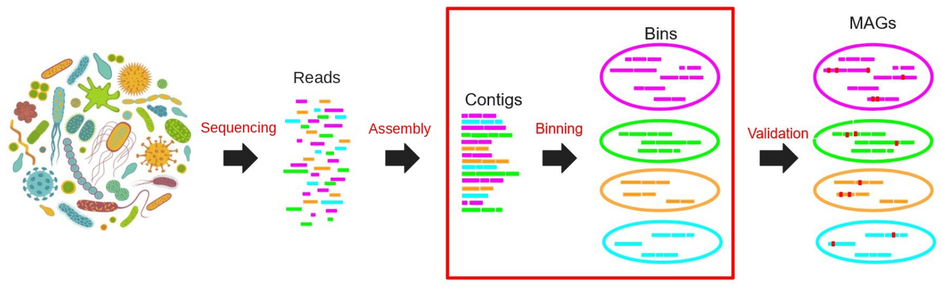


Figure 1: MAG reconstruction process

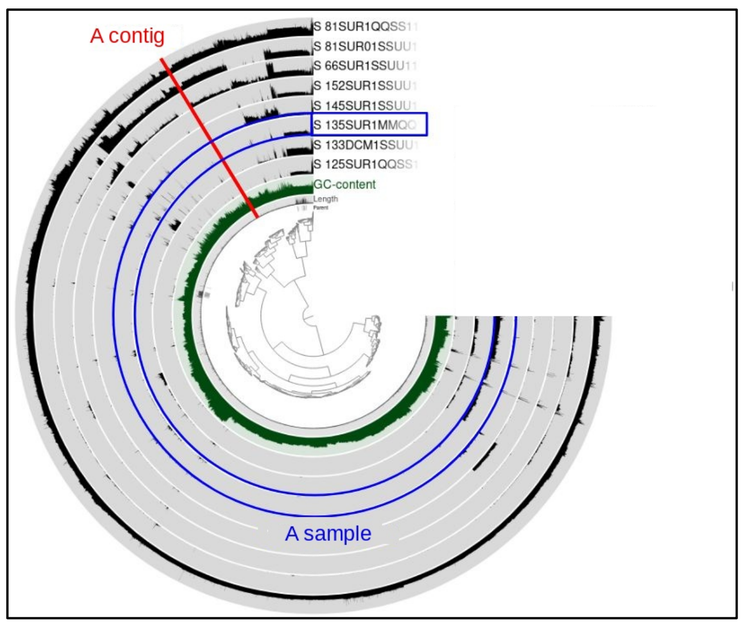


Figure 2: contig visualization with Anvi'o

RESULTS

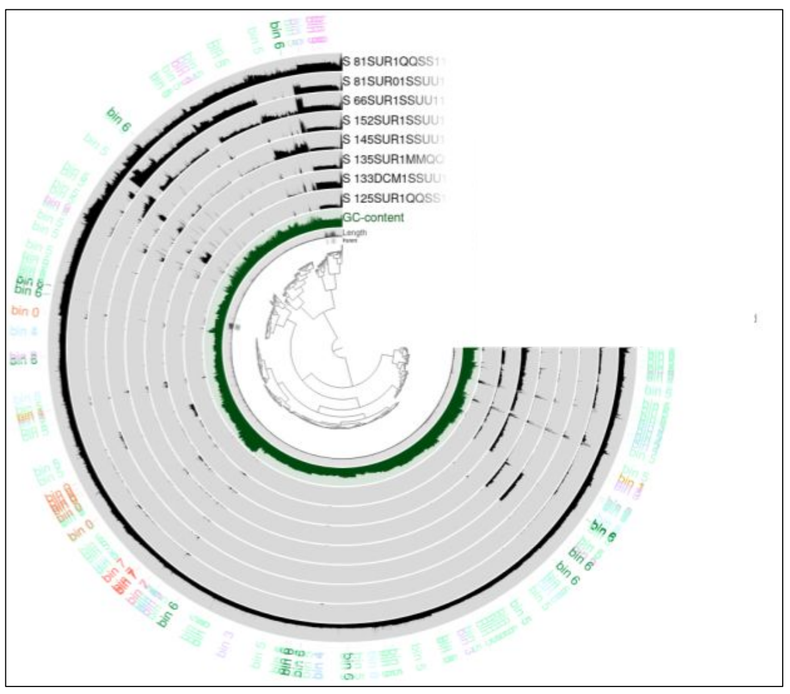




Figure 5: Cluster visualization in Anvi'o

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